

Homework

Green Machines + Metallic materials · Year 9 Science

about 80 minutes · A / M / E

Do Part 1 and Part 2 on two different days before the exam (about 40 minutes each).

Part 1 · Green Machines: Section A

1 Write the meaning of each word. [A]

(a) germination

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(b) transpiration

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(c) zygote

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2 Complete the sentences about flower parts. [A]

The _____ protects the flower while it is a bud. The _____ makes sweet nectar. The _____ is sticky so that pollen grains stay on it.

3 Name three things a seedling competes with its parent plant for. [A]

1. _____ 2. _____ 3. _____

Part 1 · Quick review

4 On a hot, dry afternoon a gardener sees the leaves of her tomato plants drooping. [A]

(a) Which process is the plant losing water by? _____

(b) Through which openings in the leaf is the water lost? _____

(c) Which tubes should be bringing water up from the roots? _____

Part 1 · Section B

5 Explain how a bee pollinates a flower. Use the names of the flower parts. [M]

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6 Explain why cross-pollination is better for a species than self-pollination. [E]

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7 Gardeners often soak bean seeds in water overnight before planting them. Explain why. [M]

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8 A carrot left in fresh water for a day becomes firm. A carrot left in very salty water becomes soft and bendy. Explain both results using osmosis. [E]

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9 A potted plant is weighed each day. All the mass it loses is water. [M]

Day	1	2	3	4
Mass (g)	500	488	476	464

(a) How much water does the plant lose each day?

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(b) Explain where the water goes.

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10 Sam explains how water gets to the leaves of a plant. [E]

Sam's answer

Water enters the root hairs by osmosis, from a low concentration of water to a high concentration of water. It travels up the phloem to the leaves. The leaves make glucose and oxygen from carbon dioxide and water. At night the plant stops respiring.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

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Also set: Green Machines booklet p.24, fill in the glossary for any ten words.

Part 2 · Metallic materials: Section A

11 Write the meaning of each word. **[A]**

(a) density

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(b) alloy

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(c) corrosion

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12 Name the heat treatment. **[A]**

(a) A blacksmith wants a horseshoe to be soft, so it is easy to shape.

(b) The edge of a chisel must be very hard.

13 Write the word equations. **[A]**

(a) lead + oxygen $\longrightarrow$ _____

(b) sodium + hydrochloric acid $\longrightarrow$ _____

Part 2 · Quick review

14 Name the property of metals that makes each one a good choice. **[A]**

(a) the copper base of a saucepan

(b) gold contacts inside a phone

(c) a steel bridge cable

Part 2 · Section B

15 A gold-coloured crown has a mass of 965 g and a volume of 50.0 cm³. The density of pure gold is 19.3 g/cm³. Is the crown pure gold? Show your working. **[M]**

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16 Aircraft bodies are made of an aluminium alloy, not steel. Use the data to explain why.

[E]

Material	Density (g/cm ³)	Strength
aluminium alloy	2.8	high
steel	7.8	very high

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17 The steel exhaust pipe of a car rusts faster than the steel body of the car. Explain why.

[M]

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18 A student adds the same mass of four metals to the same volume of dilute acid and measures the temperature rise.

[E]

Metal	magnesium	zinc	iron	copper
Temperature rise (°C)	18	7	3	0

(a) Put the metals in order of reactivity, most reactive first.

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(b) Use the data to explain your order, and name two things the student kept the same.

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19 Sam writes about alloys and reactions.

[E]

Sam's answer

1. Steel is harder than pure iron because all the atoms in steel are the same size, so the layers slide easily.
2. magnesium + hydrochloric acid makes magnesium chloride + water
3. Bronze is an alloy, so it is a mixture of metals.
4. When copper is added to dilute hydrochloric acid, bubbles of hydrogen are given off.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

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Also set: Metals booklet p.10–11 (draw the solder graph, then answer Q1–2) and the crossword on p.16.